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Data Mining and Machine Learning Applications for Educational Big Data in the University

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***Abstract*— In this paper data mining and machine learning applications for educational big data in the university are examined. We proposed prediction models in learning analytics using the student teaching data and show the effectiveness of machine learning methods. Some examples of practical applications will be shown to verify practical effectiveness.**

Keywords—learning analytics, machine learning, educational data mining, data analysis, educational big data

I. Introduction

Effective use of large amounts of data obtained by the progress of information technology is an important issue to improve management, student support, quality of education, etc. in the university. Also, in recent years, research on learning analytics that supports education and learning through data analysis has become important [1].

In this research, we consider the application of data mining and machine learning to educational big data in university. The increase of graduation postponement or dropout is a serious problem and so we propose methods that are useful for reducing them by predicting them earlier. We collect and integrate the student teaching data such as the student's grades and the class attendance situation, and further facilitate analysis by creating a database for it. The trend is analyzed by visualizing these educational big data, and the features are extracted by data mining.

We examine the prediction models in learning analytics by applying the method of machine learning. We intend to predict students who are likely to drop out of school or postpone graduation and also predict in real time students who are likely to have poor grades due to their daily low class attendance, and consider early methods to prevent them from continuing such behavior. The application of machine learning methods to such predictive models has been attempted in [2].

The purpose of this research is to analyze in more detail, and to clarify what kind of features and methods are most effective with the availability of increasing data. Furthermore we evaluate prediction performance by applying various features and methods, and clarify the application effect of machine learning methods. In addition, we will also consider the recommendations of course subjects by performing prediction of pass / fail of subject and association analysis between subjects. In this way, the purpose is to support comprehensive learning with educational big data.

II. Educational Big Data in University

A. Utilization of Internal Data at University

The progress of computerization has made it possible to collect and use various data in universities. Along with this, the approach to IR (Institutional Research) at universities has become important. IR at university refers to a wide range of activities such as collecting and analyzing internal data, formulating improvement measures, and executing and verifying measures to improve university management, student support, and the quality of education.

Main items of IR activities:

1. Survey of applicants for admission, employment situation survey, analysis of grades and GPA, factor analysis such as absence from school, retention, dropout etc., achievement survey of students.
2. Evaluation survey of university education by students, analysis of class evaluation by students, survey of graduates, etc.

As a result of conducting a survey of needs for such IR, the following items can be listed as content that we would like to work with using data.

(1) Quality assurance of education
(2) Student support and career formation
(3) Entrance examination strategy

B. Student Teaching Data

Information systems have progressed, and unified campus data is being managed in a centralized manner through them. Teaching data such as student score and attendance are also included, and recently the attendance management system by IC card-reader has been introduced. Therefore, in this research, we aim to utilize these teaching data to help students prevent their dropout and graduation postponement. The following information can be obtained for each student as teaching data such as grades, courses, class attendance, etc.

(1) Basic information

Status of Enrollment: Date of enrollment, expected date of graduation, leave of absence, course withdrawal etc.

Number of registered and acquired units, GPA (Grade Point Average) in each semester.

(2) Course information

Subjects registered for the course in each semester. The subjects also include information such as the teacher in charge, the number of credits, compulsory or elective subject.

(3) Grade information

Evaluation of subjects registered for the course in each semester. Evaluation of the subject was categorized into five levels; namely, S (extremely excellent), A (excellent), B (good), C (passed), D (fail). Units cannot be acquired when evaluation is fail. Moreover, a score (S: 4, D: 0 etc.) is given

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to these evaluations, and it is set as GP (Grade Point). The average of all subjects' GP is called GPA. GPA corresponds to overall score.

(4) Attendance information

• Number of class attendance in subjects enrolled in each semester. Average class attendance rate per semester.

• Detailed information on attendance: status of attendance for each lecture of each registered subject. It is updated daily by attendance management system by IC card.

C. *Database Construction*

In order to make it easy to analyze the data of many students, we first integrate the Excel files and put them into a database that can be used easily. Furthermore, in consideration of personal information, it is anonymized so that individual students cannot be identified.

Fig. 1 shows the flow of creating the database. First, we integrate the Excel files of individual student's teaching data for all students. In addition, information on subjects offered in each semester, student's course status, grades, attendance status, etc. was extracted and made into a database. The basic tables are described as follows:

(1) Student table

It consists of basic information such as the enrollment status of each student and credits earned per year.

(2) Subject table

Represents the subjects offered in each semester, and is composed of information such as subject name, instructor in charge, number of credits, required and selected classes.

(3) Course table

It shows the status of student's grades, and is composed of the subjects that each student has registered for each semester and the information of their evaluation.

(4) Attendance table

It shows the attendance status of students, and is composed of information on the number of attendances of subjects registered by each student in each semester and information showing the attendance status of each lecture in more detail. The attendance information detail table corresponds to all lectures of all subjects taken by all students and is the largest data.

With these basic tables, all information related to academic affairs, such as student grades and attendance, can be expressed. Therefore, it is possible to create various data from these basic tables as needed. The configuration of the database is shown in Fig2. For example, it is possible to obtain data such as the number of students, grades, passing rates, attendance rates, etc. for each subject each year, and it is also possible to obtain other information about the students. It can be used for various analysis.

III. Learning Support by Teaching Data Analysis

A. *Study Support for Low Achieving Students*

In recent years, the increase in students' dropout or graduation postponement is a problem in particular. Therefore, by analyzing the teaching data, we investigated a learning support method to prevent dropout and graduation postponement. We

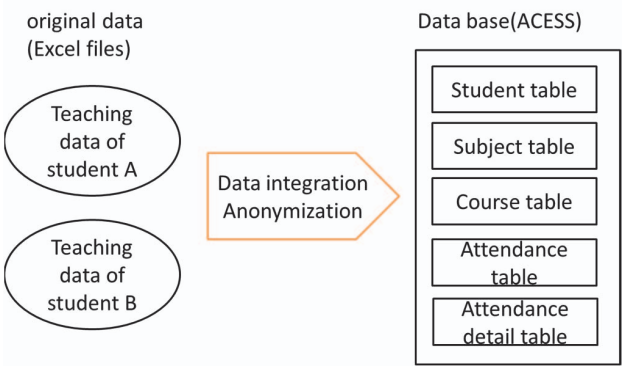

Fig.1. Construction of the database

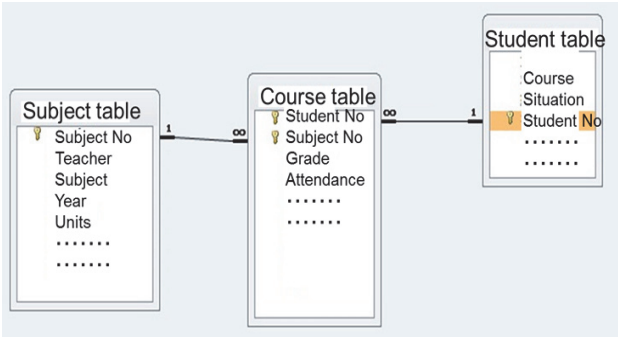

Fig.2. Database configuration

used visualization and decision tree analysis on the database.

(1) Visualization

The relationship between the number of credit units earned and the attendance rate in the first semester is shown in Fig.3. The graph shows a strong correlation between the attendance rate and the number of units to be acquired, and likewise shows the distribution of regular graduation, dropout/graduation postponement.

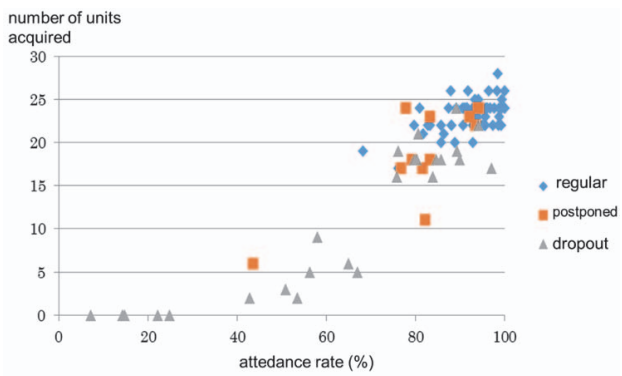

Fig. 3. The attendance rate and the number of acquisition units

In this way, by visualizing and analyzing the number of earned units and attendance rates of regular graduation, postponed graduation, and dropout students in each semester, we can analyze the trend and extract quantitative data to provide them learning support.

In addition, by organizing the results of each student into

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a database, the passing rate of each subject can also be easily determined. The passing rate of each subject is shown in Fig.4. By visualizing this way, it is possible to help select the course subjects.

(2) Analysis by decision tree

In order to examine the data in more detail, we tried to analyze it using decision tree as used in data mining. Decision tree analysis algorithm CART was applied to create a decision tree. The CART algorithm [3] is a divide-and-conquer algorithm that calculates Gini coefficient for each leaf node under arbitrary branching conditions, and uses the one with the largest difference in Gini coefficient from the parent node as the branching condition. It grows until there are no good branches, and then create a binary tree.

Fig. 5 shows an example of decision tree analysis at the end of the 1st year. For students with an acquisition rate of 37 or less and an attendance rate of 79% or less, 24 of 25 people drop out of school or postpone graduation.

B. *Construction of Learning Support Method*

Fig.6 shows the composition of the learning support method by educational data analysis. In order to reduce dropout and graduation postponement, we predict and guide students who are likely to drop out or postpone graduation early and support their preparation of study plans.

It is also important to quickly detect changes in class attendance through the daily real-time attendance status to prevent failing grades from occurring. Therefore, we also studied a method to detect early poor performance by performing weekly prediction using the average attendance rate of all subjects.

In addition, it is also important to support a plan for students with poor grades by predicting the passing or fail of the course subjects. The relationship between subjects is useful not only to predict success or failure, but also to support a course plan suitable for each student.

Therefore, we also considered association analysis between subjects. By analyzing the subjects that each student is taking at the same time and their grades, we will clarify the relationship between the subjects and support the study plan according to each student's purpose.

C. *Learning Analytics*

Learning analytics is the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs. A related field is educational data mining [4].

The process of learning analytics consists of the following four iterations.

(1) Accumulation of education and learning data

(2) Data analysis

(3) Feedback of analysis result to learner and teacher

(4) Evaluate and improve feedback effect

Learning analytics research includes behavioral prediction: Based on a large amount of data accumulated in education and learning activities, machine learning is used to create a prediction model to predict learner's behavior such as performance, dropout, etc. [5]

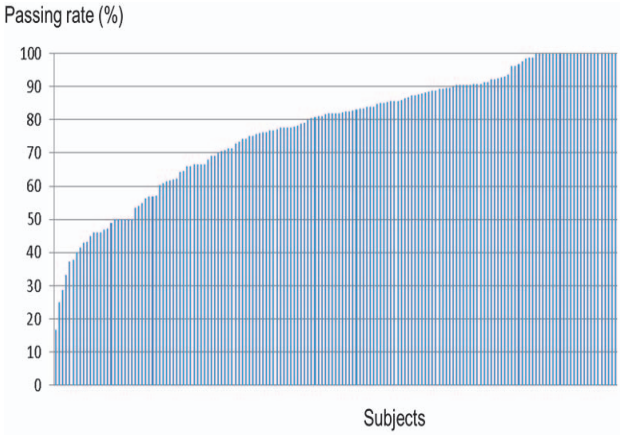

Fig.4. Passing rate of each subject.

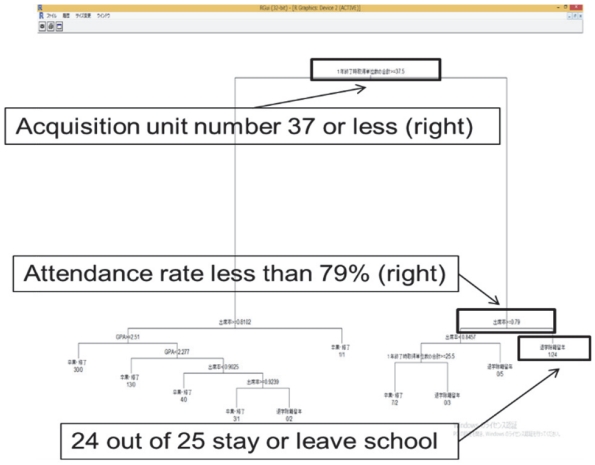

Fig. 5. Decision tree analysis at the end of the 1st year.

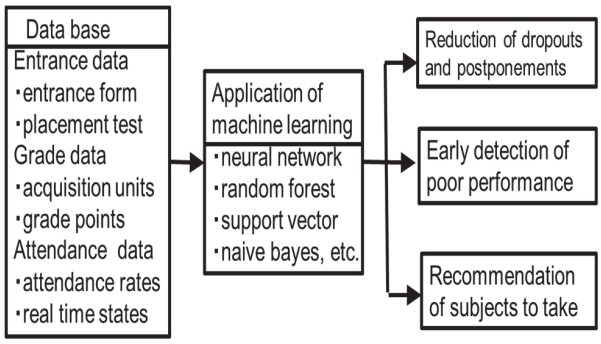

Fig.6. Learning support methods by student data analysis.

IV. Machine Learning for Prediction Models

Machine learning methods [6] [7] are applied to prevent dropouts and graduation postponements by analyzing data such as the student's achievements and attendance situation. This is aimed to clarify what kind of features and methods are most effective.

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A. Prediction of Dropout and Graduation Postponement

It is important to anticipate early and prevent students who are likely to dropout or postpone graduation. Therefore, by applying the machine learning method at each time point, how much can be predicted was studied. We used functions of Mathematica [8] and the following classification methods were used.

Classification methods:

- neural network
- logistic regression
- support vector machine
- random forest
- naïve bayes
- nearest neighbor

At the time of student's entrance to the university, we predicted using the type of entrance examination used (e.g., AO (Admission Office), through school's recommendation, general) and placement test results. In general, it is important to evaluate the predicted performance in machine learning by the F value which is the harmonic mean of the recall ratio and the precision. Although the result of naive Bayes is somewhat good, in any case it is difficult to predict at this point.

Next, at the end of the first semester, the unit acquisition rate and the average attendance rate were additionally added as features. Even here, Naive Bayes is somewhat good. However, as compared with the time of entrance, overall performance has been largely improved, and the difference is great depending on the timing and features.

Furthermore, at the end of the year, we predicted using the unit acquisition rate, average attendance rate, GPA of each semester. The results are shown in Table 1. Classes are 0 for dropout/graduation postponement and 1 for regular graduation. In general, good results have been obtained for naive Bayes, but the difference by the methods is not so large. At the end of the year, we can predict 70% or more of dropout / graduation postponement students.

Table 2 shows the prediction results at each point in time from entrance to the end of the 2nd year. As time goes on, the prediction accuracy improves and more than 80% can be predicted by the end of 2 years.

In addition, according to the rules of this department, graduation thesis studies cannot be started if the student have not acquired 100 credits or more by the end of the third year. In fact, there are quite a number of students who have to postpone their graduation because of this. Therefore, we also wanted to predict the number of units acquired until the end of the third year. The prediction results on the number of units acquired and prediction results of dropout and postponement of graduation by each method is shown in Table 3.

As described above, it was possible to predict almost accurately by comprehensively judging a plurality of methods and indicators, and it was found that it was sufficiently effective for early prediction/prevention of dropout and graduation postponement. Although about 10% of students cannot be detected by these predictions, we examined the accumulated information (including individual interviews) of the teaching system for these students in detail and found that most of them had individual circumstances such as ability, individuality and environmental change.

Table 1 Predicted results at the second semester ends.

methods	accuracy	class	recall	precision	F-score
Neural Network	0.797	0	0.662	0.783	0.718
		1	0.883	0.804	0.842
Logistic Regression	0.816	0	0.669	0.826	0.739
		1	0.91	0.812	0.858
SupportVector Machine	0.811	0	0.669	0.812	0.733
		1	0.901	0.81	0.854
Random Forest	0.811	0	0.648	0.829	0.727
		1	0.915	0.803	0.855
Naive Bayes	0.814	0	0.725	0.78	0.752
		1	0.87	0.833	0.851
Nearest Neighbors	0.797	0	0.606	0.827	0.699
		1	0.919	0.785	0.847

Table 2 Predicted results at each period in the 2 years.

period	accuracy	class	recall	precision	F-score
entrance	0.604	0	0.261	0.481	0.338
		1	0.821	0.637	0.717
half a year later	0.808	0	0.676	0.576	0.453
		1	0.892	0.675	0.743
1 year later	0.814	0	0.725	0.78	0.752
		1	0.87	0.833	0.851
1.5 years later	0.851	0	0.707	0.87	0.78
		1	0.937	0.843	0.887
2 years later	0.887	0	0.752	0.924	0.829
		1	0.964	0.870	0.915

Table 3 Predicted results of each method and number of units.

Student number	Number of units	NN	LR	SV	RF	NB	NE
9001	111	1	1	1	1	1	1
9002	21	0	0	0	0	0	0
9003	96	1	0	0	0	0	1
9004	110	1	1	1	1	1	1
9005	116	1	1	1	1	1	1
9006	14	0	0	0	0	0	0
9007	41	0	0	0	0	0	0
9008	101	1	1	1	1	1	1
9009	116	1	1	1	1	1	1
9010	77	0	0	0	0	0	0
9011	70	0	0	0	0	0	0
9012	35	0	0	0	0	0	0
9013	94	1	1	1	1	1	1
9014	108	1	1	1	1	1	1

B. Predictive Results based on Attendance Status

We also studied methods for predicting students who are likely to obtain poor grades based on daily attendance status and to prevent them early. Students with less than 10 units of credits in each semester are worrisome because that they may

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dropout or overstay. Therefore, we categorize the number of acquired units as follows: 10 units or less as poor performance (I), 10 to 19 units as median (II), 20 units or more as good performance (III).

Because attendance is managed by IC card, we can determine the attendance status of each student in each day in real time. Therefore, we predicted performance by using the average attendance rate of all subjects every week, and also examined the method of predicting students with poor performance.

The classes of each subject will be held 15 times (15 weeks) per semester and it is important to predict poor performance earlier. As an example, we examined how well we could predict the poor performance student at the sixth week when they can be still restored. The following regression methods were used

Regression methods:

- neural network
- linear regression
- random forest
- nearest neighbor

As the features, we used the average attendance rate of all subjects (weekly unit) for each week and the total cumulative average attendance rate (6 weeks) as shown in Fig.7.

At the beginning of the first semester, students have just entered the university and are attending relatively seriously. There are many students who become unable to attend as time goes on and cannot acquire credits, but they are still unpredictable at this point. We evaluate by F value. In this example, the random forest is relatively good, but in any case the accuracy is not very good.

Similarly, the predicted results at the end of the sixth week of the second semester are shown in Table 4. It is a mixed matrix of prediction results of each class (I is poor performance), the column corresponds to the actual result, and the row corresponds to the prediction result.

This has been improved considerably. Since students' attendance tendencies are almost fixed in the second semester, students with poor grades can also be predicted early. Although it is somewhat improved by using information such as the unit acquisition rate and the average attendance rate in the previous semester, it can be seen that the current attendance status is the most important.

Fig.8 shows a scatter plot of the unit acquisition rate results and the prediction results at the end of the sixth week in the second semester. For reference, the prediction results at the end of the 15th week when classes finally ended are shown in Table 5. It is the prediction result by the neural network and they can be predicted with accuracy of about 85% in both semesters. The remaining errors are considered to be due to the fact that there are aspects that cannot be predicted by attendance alone and situations that are difficult to predict.

The predicted results were compared every 3 weeks after the start of classes in both semesters. Because it aims at early detection of bad results, we use the transition of the F value of the prediction result of class I (poor performance). The prediction accuracy is not very good at the beginning of the first semester, because there are students who attend seriously at first but will not come later. On the other hand, the trend is stable in the second semester, and the prediction accuracy is good from a relatively early point because the previous semester's results and attendance data are also used.

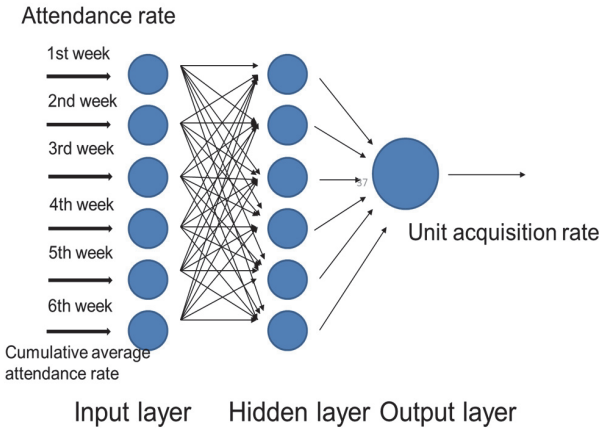

Fig.7. Construction of neural network of prediction model.

Table 4 Predicted results in the 6th week of the second semester.

methods	class	I	II	III	precision	F-score
Neural Network	I	47	10	1	0.81	0.764
	II	15	75	47	0.547	0.568
	III	3	42	134	0.749	0.742
	recall	0.723	0.591	0.736		
Linear Regression	I	40	5	1	0.87	0.721
	II	24	81	53	0.513	0.568
	III	1	41	128	0.753	0.727
	recall	0.615	0.638	0.703		
Nearest Neighbors	I	43	5	1	0.878	0.754
	II	20	71	41	0.538	0.548
	III	2	51	140	0.725	0.747
	recall	0.662	0.559	0.769		
Random Forest	I	45	10	2	0.789	0.738
	II	19	70	52	0.496	0.522
	III	1	47	128	0.727	0.715
	recall	0.692	0.551	0.703		

Table 5 Predicted results at the end of the semester.

semester	class	I	II	III	precision	F-score
first semester	I	24	3	0	0.889	0.842
	II	6	31	42	0.392	0.443
	III	0	27	241	0.899	0.875
	recall	0.8	0.508	0.852		
second semester	I	54	7	0	0.885	0.857
	II	10	82	52	0.569	0.605
	III	1	38	130	0.769	0.741
	recall	0.83	0.646	0.714		

C. Recommendation of Elective Subjects to Take

(1) Predicting pass / fail of subjects

Underperforming students often take courses that are difficult to earn credits, resulting in a shortage of credits. Therefore, we examined a method to predict the passing or fail of the elective subject for each student using neural network.

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Table 6 shows an example of the prediction result. Both predicted pass / fail with accuracy higher than the average pass rate, showing effectiveness. In addition, it was shown that accurate prediction can be made by using the results of related subjects, etc. In general, the results differ depending on the subject, and the causality is not yet clear, and it is necessary to further study in what conditions to predict.

(2) Association analysis between subjects

The relationship between students is also useful to support the study plan suitable for each student. Therefore, we also considered association analysis between subjects. By analyzing the subjects and their grades taken simultaneously by each student, the purpose is to clarify the relationship between the subjects and to use it for course support.

Fig. 9 shows an application example of the Apriori algorithm [9] for extracting association rules between subjects. This makes it possible to extract closely related subjects according to the purpose by calculating values such as confidence, lift and support.

V. Conclusion

We examined the application of data mining and machine learning techniques to educational big data in university. A variety of applications were made possible by creating a database of teaching data accumulated for each student. We proposed the learning support methods of the student by these teaching affairs data analysis.

The machine learning methods were applied to the dropout and graduation-postponement prediction as prediction models in learning analytics. We have clarified to some extent what kind of features and methods would be most effective. Furthermore, we have constructed a practical method by comprehensively judging these plural methods and indicators.

In addition, it may be possible to respond appropriately to the situation of each student by quantitatively utilizing the prediction results by machine learning as a standard trend. For future work, we will also study methods to improve the prediction performance by extracting useful features automatically from various data through deep learning. Furthermore, it is important to support student studies comprehensively by providing support for their study plans.

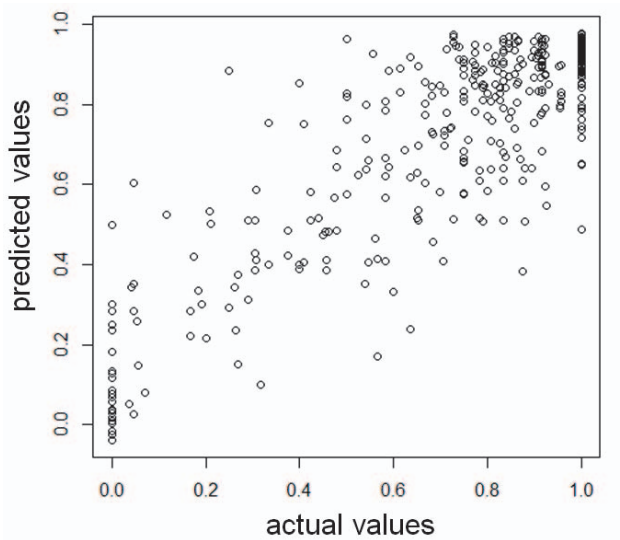

Fig 8. Predicted results in the 6th week of the second semester.

Table 6 Prediction result of success or failure.

Subject	Learning data	Predictive value
A	GPA, units, attendance rate	68%
B	GPA, units, attendance rate	73%
C	GPA, units, attendance rate, related subject grades	95%
D	GPA, number of cumulative units, attendance rate	79%

lhs		rhs	support	confidence	lift	count
[1] {3,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[2] {3,6,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[3] {2,3,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[4] {3,7,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[5] {2,3,6,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[6] {3,6,7,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[7] {2,3,7,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26
[8] {2,3,6,7,8,9}	=>	{5}	0.3939394	0.9629630	1.2970522	26

Fig 9. Application example of Apriori algorithm.

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